

**HALIFAX INFIRMARY
EXPANSION****23 65 16
COOLING TOWERS –
INDUCED DRAFT****PART 1 - GENERAL****1.1. SUBMITTALS**

- 1.1.1. Shop Drawings/Product Data: Submit cooling tower Shop Drawings/product Design Data sheets, complete with piping and power and control wiring schematics, accessories, rated capacities, weights, and all other data to confirm that the cooling tower proposed meets all requirement of the Project Agreement.

1.2. QUALITY ASSURANCE

- 1.2.1. Codes and Regulations: The cooling tower(s) and installation of cooling tower(s) are to be in accordance with requirements of the following:
- 1.2.1.1. Applicable Law;
 - 1.2.1.2. Cooling Technology Institute CTI STD-201, Certified Cooling Towers;
 - 1.2.1.3. Cooling Technology Institute CTI STD-105, Acceptance Test Code;
 - 1.2.1.4. ASHRAE Standard 90.1, Energy Standard for Buildings.

PART 2 - PRODUCTS**2.1. COOLING TOWER**

- 2.1.1. Induced draft, cross-flow, vertical air discharge, factory assembled, with thermal performance certified by the Cooling Technology Institute and energy efficiency meeting requirements of ASHRAE 90.1.
- 2.1.2. General Re: Tower Construction: The tower and all its components are to be designed to withstand all applicable wind, seismic, shipping and hoisting loads, and, unless otherwise specified, all metal components, including the support structure, casing, fan cylinder extensions, and all nuts, bolts and fasteners are to be constructed of G-235 hot-dipped galvanized steel with all cut edges given a heavy factory coat of zinc-rich compound. The tower structure is to be constructed using double-brake flanges with water-tight joints. The fan(s) and drive system, including the motor are to be factory mounted and aligned and located in the dry entering air stream. The heat transfer section(s) consist of fill, spray water distribution system, and drift eliminators.
- 2.1.3. Corrosion Resistant Coating: All mechanical equipment components exposed to the ambient air shall be provided with a corrosion resistant coating. Coating material and coating process shall have passed a minimum of 4500-hour salt spray test in accordance with ASTM standard B117.85.
- 2.1.4. Fill & Drift Eliminators: Formed from self-extinguishing PVC with a flame spread rating of five as per ASTM E84, with a spacing between fill sheets of 20 mm (¾"), shall limit drift loss to a maximum of 0.005 % of the recirculation rate and suspended from steel tubing elevated above the cold-water basin.
- 2.1.5. Water Distribution System: The water distribution system is to distribute water evenly over the fill and is to consist of an atmospheric basin equipped with removable stainless-steel covers and low flow weirs. Provide independent connections (make up water connection, overflow connection, condenser water inlet connection, and condenser water outlet connection, equalizer line connection etc.) for each cooling tower cell.

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- 2.1.6. Cold Water Basin: Sloped for positive drainage to a depressed section into which accumulated silt can be flushed to prevent the build-up of silt and debris under the fill area, a large area lift-out galvanized steel suction strainer with perforated openings sized smaller than spray nozzle orifices and a galvanized steel anti-vortexing device to prevent air entrainment, a corrosion resistant slow-acting water make-up valve with polystyrene filled plastic float for adjustment of the operating water level, circular access doors for access to the strainer and water make-up valve assembly, and all required pipe connections. The basin is to be factory lined.
- 2.1.7. Fan(s): Axial flow type, dynamically balanced with aluminium alloy blades and air discharge through a fan cylinder designed for streamlined air entry and minimum tip clearance, and equipped with a conical, non-sagging removable fan guard. Each fan and shaft is to be supported by grease lubricated, self-aligning, heavy-duty L10 (80,000 hr. life) bearings with moisture-proof seals, integral slinger collars, and lubrication nipples.
- 2.1.8. Motor(s) and Drive: TEFC motor, variable torque, specially insulated for exterior cooling tower duty, and gearbox with tire type close coupling, and guard in accordance with requirements specified in the mechanical specification section entitled Basic Mechanical Materials and Methods.
- 2.1.9. Variable Speed Drive: In accordance with requirements specified in the mechanical specification section entitled Variable Frequency Drives and supplied with the tower by the tower Supplier. The VFD is to:
- 2.1.9.1. Include a solid state, PID temperature controller to adjust frequency output of the drive in response to the tower cold water temperature, and the temperature of the cold water is to be displayed on the VFD door;
 - 2.1.9.2. Be equipped with a pipe mounting temperature sensor with well supplied loose for field installation;
 - 2.1.9.3. Be equipped with a bypass contactor to be cycled on and off while operating in bypass to maintain the set-point temperature of the cold water;
 - 2.1.9.4. Be equipped with operator controls mounted on the front of the enclosure and consisting of start and stop push button, bypass/VFD selector switch, Auto/Manual selector switch, manual speed potentiometer, and the solid-state temperature controller;
 - 2.1.9.5. Cycle the motor on/off when the minimum allowable motor speed is reached to prevent motor overheating problems.
- 2.1.10. Power and Control Panel: Factory prewired panel to control operation of the fan motor, consisting of an NEMA 4 surface wall mounting enclosure equipped with:
- 2.1.10.1. A door interlock type fused disconnect switch, lockable in the "OFF" position, and a system "ON" indicating LED;
 - 2.1.10.2. A fused control circuit transformer;
 - 2.1.10.3. An across-the-line protected type magnetic contactor for each motor speed;
 - 2.1.10.4. An enclosure mounted "HAND-OFF-AUTO" selector switch;
 - 2.1.10.5. A multiport terminal block for connection of auxiliary equipment, including a building automation system;
 - 2.1.10.6. A separately enclosed thermostatic temperature controller, with a pipe mounting temperature sensor with well supplied loose for field installation.

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- 2.1.11. Access Door Platform: Galvanized steel bar grating access door platform supported by a galvanized steel framework attached to the tower, located at the end wall access door at each end of the tower and complete with handrail, knee-rail, and toe-board.
- 2.1.12. Plenum Walkway: Factory installed galvanized steel bar walkway supported to the tower framework and extending from one access door to the opposite access door at or above the cold-water basin overflow level.
- 2.1.13. Access Ladder and Handrail: One access ladder should allow for required access to the top of two adjacent tower cells. Handrail assembly to allow access to the entire top surface of both adjacent tower cells. Handrail, knee-rail and toe-board around the perimeter of the top of the tower, designed and constructed in accordance with Nova Scotia Occupational Health and Safety Act and capable of withstanding a 890 N concentrated live load in any direction, and a vertical 450 mm (18") wide aluminium ladder with 75 mm (3") I-beam side rails and 32 mm (1¼") diameter rungs permanently attached to the tower casing and extended up from the base of the tower to the top of the handrail.
- 2.1.14. Ladder Extension: Constructed as for the access ladder, and of a length to extend from the roof or grade (as applicable) to the base of the tower for connection to the access ladder.
- 2.1.15. Ladder Safety Cage: Galvanized steel cage in accordance with Nova Scotia Occupational Health and Safety extending from a point 2.134 m (7') above the foot of the access ladder to the top of the fan deck handrail.
- 2.1.16. Pan Heater: One or more stainless steel electric immersion heaters installed in threaded couplings and secured in place in the side of the cold-water basin, and capable of maintaining 4.4 deg. C (40 deg. F) water temperature in the basin with a -26 deg. C (-15 deg. F) ambient temperature and a wind velocity of 25 km/h (15.5 miles/hr). The heater assembly is to be complete with a control probe located in the basin to monitor water level and temperature, and a wall mounting power and control panel equipped with:
 - 2.1.16.1. An NEMA 4 enclosure with nameplate;
 - 2.1.16.2. A door mounted unfused disconnect with handle lockable in the "OFF" position;
 - 2.1.16.3. A protected magnetic contactor to energize the heaters;
 - 2.1.16.4. A 24-volt secondary fused control transformer;
 - 2.1.16.5. A solid-state circuit board for temperature and low water cut-off control.
- 2.1.17. Vibration Limit Switch: Single pole, double throw, manual remote reset, adjustable vibration limit switch with contact for building automation system monitoring in a NEMA 4 housing, installed on the mechanical equipment support structure for site connection to the tower control panel to shut-down the tower should excessive vibration occur. Vibration Control System shall include:
 - 2.1.17.1. Vibration cut out switch with a weather proof cover to detect vibration
 - 2.1.17.2. Alarm indication
 - 2.1.17.3. Shut down of the cooling tower fan
 - 2.1.17.4. Time delay feature (for start up sequence)
 - 2.1.17.5. Local reset
 - 2.1.17.6. Control relay to connect to a BAS. Vibration alarms and shutdown shall be visible on the BAS

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- 2.1.18. Intentionally Deleted
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- 2.1.20. Vibration Isolation: Weatherproof vibration isolation rails, zinc coated after fabrication, designed for a 25 mm (1") static deflection and a maximum wind speed of 80 km/h (50 miles/hr), factory supplied loose with the tower and conforming to requirements specified in the mechanical specification section entitled Mechanical Vibration Control.
- 2.1.20.1. Provide a side stream condenser water filter package complete with filter backwash drain.
- 2.1.20.2. Seismic Restraint Anchoring: Suitable anchoring connection hardware factory installed on the tower to suit requirements of the mechanical specification section entitled Seismic Control and Restraint.
- 2.1.21. Sound levels shall be independently verified by a CTI-licensed sound test agency to ensure validity and reliability of the Supplier's published values. Measurement and analysis of the sound levels shall be conducted by a certified Professional Engineer in Acoustical Engineering. Sound pressure levels shall be measured and recorded in the acoustic near-field and far-field locations using ANSI S1.4 Type 1 precision instrumentation and in full conformance with CTI ATC-128 test code published by the Cooling Technology Institute (CTI). All low sound options shall be CTI certified for thermal performance.
- 2.1.22. The tower shall be capable of cooling water in accordance with the equipment schedules, and its thermal rating shall be Certified by the Cooling Technology Institute.
- 2.1.23. The tower shall be capable of a minimum gpm/Hp efficiency per ASHRAE Standard 90.1.

PART 3 - EXECUTION**3.1. Installation**

- 3.1.1. Provide vibration isolation (equipment isolators, flexible pipe connectors, etc.) in accordance with Section 20 05 20 – MECHANICAL VIBRATION CONTROL.
- 3.1.2. Provide Variable Frequency Drive(s) (VFD) for each cooling tower fan in accordance with Section 20 05 30 - VARIABLE FREQUENCY DRIVES.
- 3.1.2.1. Provide start-up and vibration testing throughout the fan speed range. Identify and lockout any natural frequency vibration levels which may exceed CTI guidelines.
- 3.1.3. Provide cooling tower support beams between the cooling tower and the vibration spring isolators where required to meet cooling tower manufacturer's requirements.
- 3.1.4. Provide isolation valves at all tower connections, including equalising lines and sweeper systems where applicable, and drain valves in accordance with Section 22 13 16 – DRAINAGE WASTE AND VENT PIPING AND VALVES. A flow balancing valve shall be installed on the piping to each hot water basin to ensure equal flow.
- 3.1.5. Provide drain valves on the low points and either side of isolation valves of the condenser water and equalizer piping to achieve complete system drainage.
- 3.1.6. Pipe connections to the cooling tower (condenser water, equalizer, etc.) shall be provided with twin-sphere flexible connections and isolation valves. The make-up water pipe connections shall have braided stainless steel flexible connections.

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- 3.1.7. Provide equalizer pipe or header between each cell basin of a multiple cell or multiple cooling tower installation. The equalizer line shall be installed complete with a capped clean-out.
- 3.1.8. Provide an adjustable bleed-off connection in the water inlet line to the tower above the sump pump level and adjust to the required bleed-off rate.
- 3.1.9. Provide passivation of the metal surfaces of the cooling towers in order to prevent the premature corrosion of the galvanizing. The Contractor shall hire the services of The building operator's prescribed chemical treatment company to complete the passivation process and any associated chemical treatment.

END OF SECTION